CrowdWisdomTrading: Quantitative Prediction Report

Multi-Horizon Sentiment & Market Price Evaluation | Generated 2026-10-19 16:09 UTC

1. Executive Summary & Free-First Architecture

This report documents out-of-sample predictive performance across 6 horizons (n+0 to n+5) using exclusively free tools, local VADER NLP sentiment analysis, and chronological walk-forward cross validation without lookahead leakage.

2. Walk-Forward Validation Metrics (Out-of-Sample)

Horizon	Model	MAE ($)	RMSE ($)	MAPE (%)	Dir Acc (%)	Base MAE ($)
n+0	Persistence	$187.2	$292.3	0.3%	49.8%	$187.2
n+0	Hist Gradient	$188.5	$294.1	0.3%	50.0%	$187.2
n+0	Random Forest	$188.7	$294.6	0.3%	49.9%	$187.2
n+0	Ridge	$192.6	$297.3	0.3%	51.5%	$187.2
n+1	Persistence	$187.2	$292.3	0.3%	49.8%	$187.2
n+1	Hist Gradient	$188.5	$294.1	0.3%	50.0%	$187.2
n+1	Random Forest	$188.7	$294.6	0.3%	49.9%	$187.2
n+1	Ridge	$192.6	$297.3	0.3%	51.5%	$187.2
n+2	Persistence	$265.8	$409.3	0.4%	50.2%	$265.8
n+2	Random Forest	$269.4	$414.3	0.4%	49.6%	$265.8
n+2	Hist Gradient	$270.0	$413.9	0.4%	50.2%	$265.8
n+2	Ridge	$277.1	$418.3	0.4%	51.2%	$265.8
n+3	Persistence	$327.0	$495.2	0.5%	49.8%	$327.0
n+3	Hist Gradient	$331.2	$502.3	0.5%	53.3%	$327.0
n+3	Random Forest	$331.9	$504.6	0.5%	50.7%	$327.0
n+3	Ridge	$341.1	$507.9	0.5%	50.6%	$327.0
n+4	Persistence	$376.3	$564.7	0.5%	50.7%	$376.3
n+4	Random Forest	$385.6	$579.5	0.5%	51.7%	$376.3

3. Quantitative Trading Strategy Performance

Model	Horizon	Cum Ret (%)	Sharpe	Sortino	Max DD (%)	Win Rate (%)	Trades
Ridge	n+1	-24.9%	-5.78	-8.06	-25.5%	40.7%	292
Random Forest	n+1	-4.2%	-1.84	-2.84	-5.8%	45.2%	94
Hist Gradient	n+1	-13.5%	-3.86	-4.94	-15.9%	45.5%	148